Final Feature List: AI Career Intelligence Platform

1. Multi-Source Candidate Profile Analysis

Users can provide:

Resume in PDF or DOCX format

GitHub profile, when relevant

Portfolio website

LinkedIn profile PDF, exported data, or pasted profile content

Target job description

Career goal and preferred role

The system combines all available sources into one reusable, structured candidate profile.

GitHub remains optional so candidates from marketing, finance, HR, sales, design, banking, operations, and other non-technical fields are not disadvantaged.

2. Career Goal Selection

Users can choose a specific objective, such as:

Find a first job

Find an internship

Improve their current professional profile

Switch careers

Transition into a particular specialization

Prepare for a target role

Move into a senior position

Move into leadership or management

Improve job-market competitiveness

All subsequent recommendations are personalized around that objective.

3. Specialization Detection

The AI identifies the candidate's actual professional specialization based on evidence from their experience, projects, skills, portfolio, and other submitted information.

Examples include:

Backend Development

Frontend Development

Full-Stack Development

QA Automation

Manual QA

Data Analysis

Data Science

AI and Machine Learning

DevOps

Cybersecurity

Product Management

Project Management

UI/UX Design

Digital Marketing

Sales

Human Resources

Recruitment

Finance

Accounting

Banking

Operations

Customer Service

Content Writing

Graphic Design

Supply Chain

The result includes:

Primary specialization

Secondary specializations

Estimated seniority level

Confidence score for each specialization

Evidence supporting each conclusion

4. Career Readiness Dashboard

The system provides a consolidated dashboard containing:

Overall career-readiness score

Resume-quality score

ATS-readiness score

Skill-alignment score

Job-match score

Interview-readiness score

Portfolio-strength score, where applicable

Profile-completeness score

Each score includes an explanation instead of appearing as an unsupported number.

5. Resume Analysis

The system evaluates:

Resume structure

Section completeness

Contact-information placement

Professional summary

Work-experience descriptions

Project descriptions

Education

Skills

Certifications

Grammar and clarity

Repetitive wording

Weak action verbs

Overly long bullets

Missing achievements

Missing measurable impact

Date inconsistencies

Role relevance

Formatting and readability concerns

It provides prioritized corrections based on importance.

6. ATS Readiness Score

The platform calculates an explainable ATS-readiness score using deterministic rules and job-specific comparisons.

The score can consider:

Required resume sections

Contact details

Relevant keywords

Required skills

Experience relevance

Job-title alignment

Bullet-point quality

Quantified achievements

Formatting and readability

Job-description compatibility

The platform clearly presents this as its own ATS Readiness Score, not as a guaranteed score from every commercial ATS.

7. Resume Bullet Rewriting

The AI rewrites weak resume bullets into stronger, action-oriented statements.

It can improve:

Clarity

Professional tone

Technical or functional detail

Achievement focus

Relevance to the target role

ATS keywords

Conciseness

The system does not invent achievements or numerical results. When a metric is unavailable, it can:

Suggest a placeholder

Ask the user for the missing value

Rewrite the bullet without fabricated numbers

8. Professional Summary Optimization

The system can create or improve:

Resume professional summary

Career objective

Short professional biography

Role-specific profile summary

The summary is based only on verified information from the candidate profile.

9. GitHub Profile Analysis

For technical candidates, the platform can analyze public GitHub information such as:

Public repositories

Programming languages

Frameworks

Repository topics

README quality

Documentation

Testing practices

Project complexity

Code organization

Dependency files

CI/CD workflows

Docker usage

Recent activity

Project diversity

Open-source contributions

GitHub popularity metrics, such as stars, are treated as supporting signals rather than primary measures of skill.

10. Portfolio Analysis

The platform can evaluate technical and non-technical portfolios.

It can examine:

Project quality

Case studies

Work samples

Project descriptions

Skills demonstrated

Problem-solving explanations

Results and impact

Visual presentation

Professional storytelling

Navigation and clarity

Missing project details

Contact and social links

For non-technical users, it can evaluate marketing campaigns, writing samples, design work, business projects, HR initiatives, sales achievements, and other relevant evidence.

11. LinkedIn Profile Analysis

Users can upload a LinkedIn profile PDF, paste profile content, or provide exported profile data.

The system evaluates:

Headline

About section

Experience descriptions

Skills section

Certifications

Featured content

Project presentation

Keyword coverage

Recruiter discoverability

Profile completeness

Alignment with the target role

The platform does not depend on unauthorized LinkedIn scraping.

12. LinkedIn Profile Optimization

The platform can generate:

Improved LinkedIn headline

Rewritten About section

Improved experience descriptions

Suggested skills

Suggested featured projects

Certification recommendations

Keyword recommendations

Profile-completeness improvements

Users can generate only the sections they need, reducing unnecessary AI calls.

13. Target Job Description Analysis

Users can:

Paste a job description

Upload a job-description document

Select a job from the platform’s internal dataset

Choose a matched job from retrieved listings

The system extracts:

Required skills

Preferred skills

Responsibilities

Experience requirements

Education requirements

Tools and technologies

Soft skills

Seniority expectations

Important ATS keywords

14. Skill Gap Analysis

The platform compares the candidate profile against a target job or career path.

Skills are categorized as:

Matched

Partially matched

Missing

Transferable

Claimed but unverified

Not relevant

For every important gap, the system explains:

Why the skill matters

How urgent the gap is

How the candidate can develop it

What project or task can demonstrate it

Whether it is essential or optional

15. Job Matching

The platform retrieves and ranks relevant jobs using structured filtering, semantic retrieval, and RAG.

Matching can consider:

Specialization

Skills

Experience

Seniority

Industry

Education

Location

Remote or on-site preference

Employment type

Career goal

Transferable skills

Each matched job includes:

Match score

Matched skills

Missing skills

Relevant experience

Potential concerns

Explanation of why the role fits

Recommendation on whether the user should apply

16. Explainable Job Recommendations

The system does not only produce a ranking.

For each job, it explains:

Which evidence supports the match

Which requirements are fully met

Which requirements are partially met

Which important requirements are missing

How competitive the candidate may be

What to improve before applying

Which resume sections should be tailored

17. Personalized Project Recommendations

The system recommends practical projects designed to close specific skill gaps.

Each project can include:

Project title

Problem statement

Target specialization

Skills demonstrated

Features to build

Recommended tools

Difficulty level

Estimated scope

Learning objectives

Suggested deliverables

Resume bullet example

Portfolio presentation advice

Connection to real job requirements

Project recommendations are available for technical and non-technical sectors.

Examples may include:

Automation framework for QA

REST API system for backend development

Campaign analytics project for marketing

Recruitment dashboard for HR

Sales pipeline analysis for sales roles

Credit-risk dashboard for finance

Customer-retention analysis for business roles

18. Personalized Career Roadmap

The platform creates a structured roadmap based on:

Current specialization

Target specialization

Career objective

Skill gaps

Available study time

Experience level

Target jobs

Available durations can include:

7-day preparation plan

30-day improvement plan

90-day roadmap

Six-month career-transition plan

The roadmap contains:

Prioritized skills

Weekly milestones

Practical exercises

Projects

Application targets

Resume updates

Portfolio updates

Interview preparation

Success criteria

19. Daily, Weekly, and Monthly Learning Plans

Users can receive a detailed schedule with:

Daily tasks

Weekly goals

Monthly milestones

Study topics

Practice activities

Projects

Review sessions

Job-application activities

Interview practice

Progress checkpoints

The schedule adapts to the user's available hours and target timeline.

20. Interview Preparation

The system generates personalized interview preparation based on:

Resume

Candidate profile

Target job

Portfolio

GitHub profile

Experience level

Skill gaps

Question categories include:

Technical or role-specific questions

Behavioral questions

Situational questions

Resume-based questions

Project-based questions

Portfolio-based questions

Leadership questions

Career-transition questions

Job-description-specific questions

21. Interactive Mock Interview

The platform can conduct a conversational mock interview.

Workflow:

AI asks a question

→ User answers

→ AI evaluates the answer

→ AI asks a relevant follow-up

→ User receives feedback

Feedback can cover:

Accuracy

Relevance

Clarity

Structure

Evidence

Technical depth

Confidence

Missing details

Suggested improvement

22. Resume Claim Verification Questions

The platform identifies important claims that an interviewer may want to verify.

Examples include:

Technologies used

Leadership claims

Project ownership

Performance improvements

Revenue or sales impact

Automation experience

Team collaboration

System architecture

Testing responsibilities

Certifications

Tools and methodologies

The platform generates targeted questions and follow-ups for each claim.

It does not claim to detect lies. Instead, it reports:

Evidence strength

Claim-verification confidence

Inconsistencies

Claims requiring clarification

Questions the candidate should be prepared to answer

23. Answer Evaluation

Users can submit answers to interview questions.

The system evaluates:

Whether the answer addresses the question

Whether examples are specific

Whether claims are consistent with the resume

Whether the STAR method is used effectively

Whether technical details are credible

Whether the answer demonstrates impact

Whether follow-up questions may expose missing knowledge

It then provides an improved answer structure without inventing experience.

24. Cover Letter Generation

The platform generates a personalized cover letter for a selected job.

It uses:

Candidate profile

Relevant experience

Skills

Achievements

Career goal

Job requirements

Company and role information

Preferred tone

Available outputs may include:

Standard cover letter

Concise cover letter

Entry-level cover letter

Career-transition cover letter

Email application version

The system does not fabricate experience or achievements.

25. Job-Specific Resume Tailoring

For a selected job, the platform can recommend:

Skills to emphasize

Bullets to reorder

Keywords to add

Summary changes

Projects to highlight

Irrelevant information to reduce

Missing evidence to add

Experience descriptions to rewrite

It can generate a job-tailored version while preserving factual accuracy.

26. Recruiter Outreach Content

The platform can generate:

LinkedIn connection message

Recruiter outreach message

Job-application email

Follow-up email

Interview thank-you email

Referral request message

Short professional introduction

All generated messages use the existing candidate profile.

27. Career Transition Analysis

For users changing industries or professions, the platform identifies:

Transferable skills

Skills that need reframing

Experience that remains relevant

Missing foundational knowledge

Suitable transitional roles

Recommended projects

Realistic learning sequence

Resume positioning strategy

Likely interview concerns

28. Evidence-Based Recommendations

Every major conclusion should show supporting evidence.

For example:

Primary specialization: QA Automation

Confidence: 84%

Evidence:

- Automated testing experience listed in the resume

- Playwright configuration found in GitHub

- API testing project shown in the portfolio

- Target roles align with test automation

This applies to:

Specialization

Skill scores

ATS feedback

Job matches

Project recommendations

Interview questions

Career-roadmap priorities

29. Universal Profession Support

The platform is designed for users across different sectors.

It should not assume that every user has:

GitHub

A technical portfolio

Programming experience

A computer-science degree

A LinkedIn profile

The evaluation changes according to the candidate's field.

For example:

Developers may be evaluated through code and repositories.

Designers may be evaluated through case studies.

Marketers may be evaluated through campaigns and analytics.

Sales professionals may be evaluated through targets and conversions.

HR professionals may be evaluated through hiring, retention, and HR operations.

Finance professionals may be evaluated through analysis, reporting, controls, and business impact.

30. Downloadable Career Report

Users can generate a report containing:

Candidate summary

Detected specializations

Confidence scores

Career-readiness scores

Resume findings

ATS analysis

Strengths

Weaknesses

Skill gaps

Job matches

Recommended projects

Career roadmap

Interview questions

LinkedIn recommendations

The report can be downloaded as a PDF.

31. Saved Analysis History

Registered users can save and reopen:

Candidate profiles

Resume analyses

Job matches

Skill-gap reports

Career roadmaps

Cover letters

LinkedIn rewrites

Interview sessions

Generated reports

Cached results prevent the system from repeating expensive AI analysis when nothing has changed.

32. Privacy and Data Controls

The platform should include:

Private document storage

User-specific access control

Temporary-file deletion

Optional deletion after parsing

“Delete my data” control

No public access to uploaded resumes

No sensitive document contents in logs

Clear indication of what data is retained

Final User Journey

1. Upload resume

2. Add optional GitHub, LinkedIn, or portfolio information

3. Select career goal and target role

4. Receive specialization and confidence analysis

5. View resume and ATS scores

6. Compare profile against target jobs

7. Review skill gaps

8. Discover recommended projects

9. Improve resume and LinkedIn profile

10. Generate job-specific cover letters

11. Practice personalized interviews

12. Follow a career and learning roadmap

13. Match with relevant jobs

14. Download or save the complete career report

The optimized project keeps the full feature set, but generates expensive content only when requested. A shared candidate profile, deterministic scoring, cached results, conditional tools, and RAG-based retrieval allow the platform to remain practical on free deployment tiers.

Recommended Optimized Free Stack

Layer	Tool
Frontend	React, Vite, TypeScript
UI	Tailwind CSS, shadcn/ui
Backend	FastAPI
Agents	OpenAI Agents SDK
Model routing	LiteLLM
Free LLM	A currently free Gemini Flash model
Database	Supabase PostgreSQL

Vector search	**Supabase pgvector**
Authentication	**Supabase Auth**
Storage	**Supabase Storage**
Resume parser	**PyMuPDF, python-docx**
GitHub	**GitHub REST API**
Portfolio extraction	**HTTPX, BeautifulSoup, Trafilatura**
Scoring	**Pure Python**
Matching	**Python + pgvector**
Frontend deployment	**Vercel**
Backend deployment	**Render**
CI/CD	**GitHub Actions**